

ZERMATT, SwitzerlandWMO: **067480**Lat: **46.0292N**Lon: **7.7525E**Elev: **5380**StdP: **12.05**Time Zone: **1.00 (EUC)**Period: **01-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	5.9	9.4	-11.7	3.6	13.5	-8.7	4.2	13.9	14.7	32.9	11.6	28.3	2.9	220	0.651

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.9	76.4	55.9	73.2	55.0	70.2	53.9	57.5	71.2	56.3	68.9	55.1	66.8	10.5	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
53.5	74.5	57.9	52.0	70.5	57.0	50.6	66.9	56.6	27.1	71.1	26.2	69.0	25.4	67.1	61.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.0	15.4	12.9	DB	2.4	81.5	4.2	2.9	-0.6	83.6	-3.1	85.2	-5.5	86.8	-8.5	88.9	(4)
(5)			WB	0.2	60.1	3.9	0.8	-2.6	60.7	-4.8	61.2	-7.0	61.6	-9.8	62.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	41.7	25.0	26.3	33.2	41.1	46.9	55.2	57.9	56.8	50.5	44.3	34.2	27.6		
	DBStd	13.41	6.77	7.93	7.18	6.41	6.35	6.38	5.63	5.97	5.78	6.66	7.56	7.22		
	HDD50	3827	775	663	520	271	135	22	7	12	63	192	473	694		
	HDD65	8534	1240	1083	985	716	560	298	228	263	436	643	923	1159		
	CDD50	790	0	0	0	5	40	179	252	222	78	14	0	0		
	CDD65	19	0	0	0	0	0	4	7	8	0	0	0	0		
	CDH74	209	0	0	0	0	2	55	76	75	1	0	0	0		
	CDH80	15	0	0	0	0	0	5	5	5	0	0	0	0		
	WSAvg	4.1	2.9	3.4	4.0	5.1	5.3	5.1	4.9	4.6	4.3	3.8	3.2	2.9		
	PrecAvg	31.7	1.7	1.7	1.9	2.4	4.0	3.1	2.7	3.0	2.7	3.3	3.1	2.2		
(15) Precipitation	PrecMax	44.6	4.2	6.0	4.1	6.9	9.0	7.1	6.1	6.3	8.7	11.8	10.3	5.2		
	PrecMin	18.1	0.3	0.0	0.4	0.1	1.5	1.3	0.9	1.0	0.7	0.8	0.3	0.1		
	PrecStd	5.8	1.1	1.3	1.0	1.8	2.1	1.2	1.0	1.3	1.9	2.3	2.4	1.3		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	44.3	49.0	55.4	63.8	71.0	79.2	80.2	80.4	71.7	64.7	54.9	45.8		
		MCWB	34.6	35.8	38.9	45.1	51.7	56.1	57.6	56.5	53.5	47.7	40.2	34.6		
	2%	DB	39.9	43.3	51.3	59.8	66.3	75.2	76.0	75.8	68.1	61.0	50.4	41.9		
		MCWB	32.5	33.2	37.3	43.0	49.3	55.3	56.0	56.0	51.9	46.2	38.9	33.5		
	5%	DB	36.7	39.4	47.7	56.1	62.4	71.3	72.9	71.8	64.5	57.7	47.1	38.7		
		MCWB	31.2	31.3	36.1	41.5	47.2	54.0	55.2	54.6	50.2	44.8	38.1	32.2		
	10%	DB	34.0	36.1	43.9	52.5	58.5	67.7	69.6	68.3	61.0	54.5	43.7	35.8		
		MCWB	29.4	29.8	34.2	39.9	45.8	52.8	54.2	53.8	48.9	44.2	37.1	30.5		
	0.4%	WB	36.9	37.1	40.8	46.2	52.8	58.3	59.2	59.0	55.1	50.5	44.0	37.3		
		MCDB	40.4	45.5	51.4	59.9	67.8	73.0	75.1	73.7	65.7	57.9	49.0	41.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	34.1	34.6	38.6	44.0	50.5	56.6	57.6	57.2	53.4	48.6	41.2	35.0		
		MCDB	38.1	41.1	48.0	56.8	62.5	70.8	71.4	69.9	63.4	55.7	46.4	39.4		
	5%	WB	32.0	32.5	36.9	42.3	48.7	55.2	56.4	55.8	51.8	47.1	39.5	33.2		
		MCDB	35.7	38.0	45.0	53.9	59.1	68.3	69.0	67.4	60.8	54.0	44.8	37.2		
	10%	WB	29.8	30.4	35.3	40.8	47.0	53.7	55.1	54.7	50.4	45.5	37.7	31.5		
		MCDB	33.7	35.2	42.6	50.8	56.3	65.6	66.7	65.3	58.8	52.4	43.4	35.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	13.5	15.0	16.4	18.0	18.0	19.4	19.9	18.8	17.7	16.9	13.4	12.8		
		MCDBR	15.2	18.1	20.8	22.7	23.8	25.0	24.4	24.2	22.4	21.1	17.2	15.0		
		MCWBR	10.5	11.3	11.2	11.1	11.0	10.7	10.0	10.2	10.3	10.5	9.6	9.3		
		MCDBR	13.5	16.5	18.4	21.1	20.8	22.4	21.7	20.9	19.1	17.1	14.7	13.5		
	5% WB	MCWBR	9.9	10.7	10.1	10.5	9.8	10.1	9.7	9.5	9.4	9.1	8.9	9.2		
(40) Clear-Sky Solar Irradiance	taub		0.240	0.267	0.315	0.362	0.379	0.386	0.381	0.368	0.345	0.318	0.274	0.238		
	taud		2.296	2.225	2.152	2.145	2.201	2.229	2.246	2.293	2.346	2.395	2.395	2.330		
	Ebn at Noon		279	290	288	283	281	279	278	276	268	262	262	267		
	Edh at Noon		24	32	41	45	44	43	42	38	33	27	22	21		
(44) All-Sky Solar Radiation	RadAvg		480	757	1119	1401	1602	1898	1908	1655	1285	826	485	397		
	RadStd		57	100	151	187	139	156	149	127	105	96	72	56		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (104 neighbors)		N/A	N/A	N/A	+1.28	+0.74	+0.56	N/A	N/A	+103	+29

Nomenclature: See separate page